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HCI & Computer Graphics

Milestone 2

Needs Finding and Initial Concept

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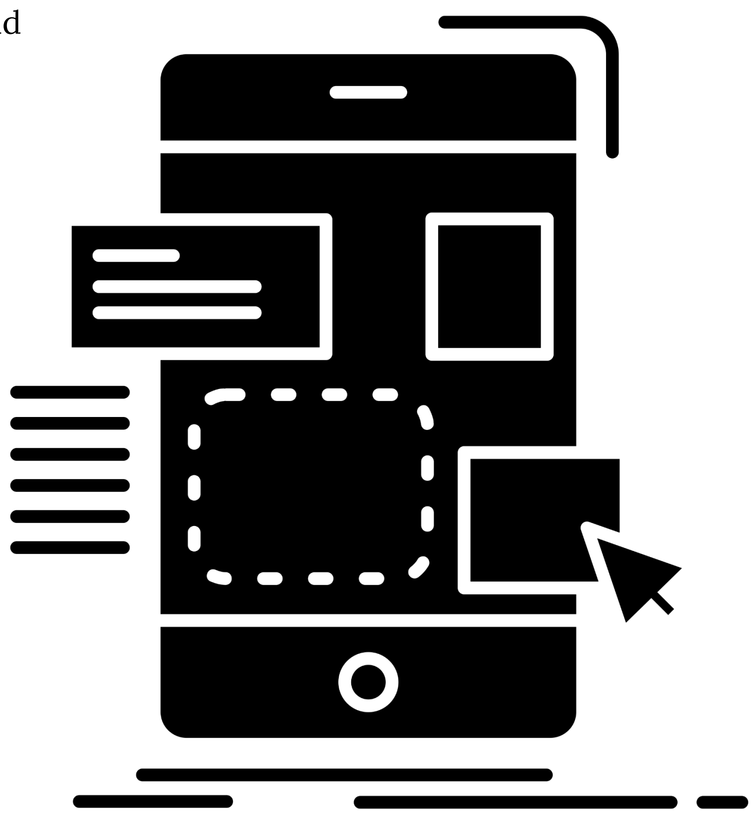

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Milestone 2: Initial Concept and Needs Finding Plans

HCI & Computer Graphics Course Project Documentation

Section 1: Needs Finding Report

1.1 Target Population

Pakistani adults (ages 18–50) managing household or personal budgets across informal and formal retail channels. Two primary groups: (A) Homemakers and household heads with low-to-moderate digital literacy who track daily expenses, and (B) university students and freelancers who handle personal budgets and have previously attempted digital finance tools.

1.2 Study Participants

Ten undergraduate students were interviewed. All participants were from BNU's CS/SE programs, Section C. Participants are anonymized below:

ID	Group	Profile	Primary Tracking Method
P1	A	Homemaker, 40s, low digital literacy	Physical notebook (Khata)
P2	A	Homemaker, 30s, moderate digital literacy	Notebook + WhatsApp notes
P3	A	Head of household, 50s, low digital literacy	Mental tracking + receipts in drawer
P4	A	Local shopkeeper, 35, moderate digital literacy	Paper ledger
P5	A	Homemaker, 28, moderate digital literacy	Combination of notes app + notebook
P6	B	University student, 21, high digital literacy	Abandoned app (Wallet); now spreadsheet
P7	B	University student, 22, high digital literacy	No consistent system
P8	B	Freelancer, 26, high digital literacy	Notion + manual receipt photos

P9	B	SME owner, 30, moderate digital literacy	WhatsApp messages to self
P10	B	University student, 23, moderate digital literacy	Abandoned Cashew app; now mental tracking

1.3 Findings: User Needs and Current Practices

Theme 1 - Receipt Handling is Chaotic

Most participants received handwritten or faded receipts that were either lost within hours or stored haphazardly. P1 kept receipts in a kitchen drawer, P3 admitted throwing most away. P8 took photos but never reviewed them. No participant had a reliable, consistent receipt workflow.

"I just put it in my purse. By the time I get home, half of them are crumpled." - P2

Theme 2 - Manual Entry is the Primary Friction Point

The single biggest reason for abandonment across both groups was the effort of manual data entry. Participants described feeling like the overhead of logging negated the benefit of tracking.

"I used Cashew for two weeks. Entering every purchase individually just felt like a second job." - P10

Theme 3 - Informal Expenses Go Unlogged

Cash-only transactions — rickshaws, local vendors, roadside stalls — were universally skipped. No existing tool accommodates expenses that produce no receipt. P7 estimated 30–40% of daily spending was never tracked.

Theme 4 - Category Systems Feel Imposed

Participants in Group A did not think in app-defined categories (food, transport, utilities). Their mental model was organized around occasions ('school run', 'weekly bazar') rather than abstract types. Group B users found default categories did not match Pakistani spending patterns (no 'rickshaw', no 'ration').

Theme 5: Trust and Verification Matter

When shown the concept of AI-based receipt scanning, 7 of 10 participants expressed skepticism about accuracy with handwritten or mixed Urdu/English text. All stated they would need an easy 'fix it' option before trusting the log.

"If it reads wrong and I don't notice, my whole month's budget is off. I need to be able to check." - P4

Functional Requirements Identified

- Camera-to-log pipeline that handles handwritten, faded, and mixed-language receipts
- Voice input for logging informal/no-receipt cash transactions
- Inline verification step after any AI-parsed entry (touch-to-edit)
- Flexible, user-defined categories that reflect local spending patterns
- Weekly/monthly summary with zero manual aggregation effort
- Offline capability, unreliable internet is common among target users

Constraints Identified

- Must not require internet for basic logging (users in low-connectivity areas)
- Avoid overwhelming onboarding, Group A users will abandon on first complexity
- Do not force categories on the user at the point of entry
- Must work on low-end Android devices (primary devices in target population)
- No subscription model, users resist recurring cost for a utility app

1.4 Implications for Design

Our original assumption, that OCR alone would suffice, was challenged. Participants require a hybrid input model: camera, voice, and manual entry must all be first-class options, not fallbacks. The verification step is non-negotiable; trust in AI output must be earned through transparency, not assumed. Additionally, the needs of Group A users pushed us toward a simpler default UI with power-user features hidden behind a toggle, rather than presenting all features upfront.

1.5 Affinity Diagramming Process

Interviews were audio-recorded and reviewed. Key observations were extracted as individual notes, approximately 180 notes across all 10 interviews. Notes were organized into clusters on a shared digital board (FigJam), first individually then as a pair. Emerging clusters were labeled and regrouped until stable themes appeared. The five themes above represent the final affinity clusters after two rounds of reorganization. Themes were then ranked by frequency and severity to prioritize functional requirements. Evidence of the process (note excerpts and cluster labels) is described below.

Affinity Cluster	Note Count	Representative Notes
Receipt Chaos	38	“Lost it on the way home”, “can’t read the number”, “don’t bother saving it”
Manual Entry Fatigue	42	“takes too long”, “gave up after a week”, “feels like extra homework”
Informal Cash Gaps	31	“rickshaw has no receipt”, “vendor just says the amount”, “no way to log this”
Category Mismatch	29	“no ration category”, “groceries doesn’t fit my spending”, “I don’t think like that”
AI Trust / Verification	40	“what if it reads wrong”, “I need to double-check”, “show me what it understood”

Section 2: Design Concept

2.1 Updated Project Brief

Target Users (refined):

Pakistani adults (18–50) managing personal or household budgets across informal and formal retail. Two tiers: low-digital-literacy homemakers who need a near-zero-friction interface, and moderate-to-high-literacy students/freelancers who need speed and data density. Both tiers share the core need: logging without manual typing.

System: DailyKhata (updated)

A multimodal AI expense logging app with three input modes, camera (receipt scan), voice ('spent 500 on vegetables'), and manual entry, with a mandatory inline verification step after any AI-parsed input. The default UI is minimal (one tap to log); a 'Pro View' reveals spreadsheet-style data management for power users.

Updated requirements:

- Multimodal input: camera, voice, manual — all equally accessible from home screen
- AI parsing with inline edit-before-save verification (replaces auto-commit)
- User-defined categories with sensible local defaults (Ration, Transport, Utility, Misc)
- Offline-first architecture with background sync
- Weekly plain-language summary (no charts by default; optional toggle)
- Lightweight onboarding: functional in under 60 seconds

Updated constraints:

- No mandatory account creation at launch
- No forced category selection, 'Misc' is always a valid bucket
- No subscription paywall on core logging features
- Must run on Android 9+ with 2GB RAM

2.2 Storyboards

Three storyboards are described below, illustrating key user flows. In the actual submission, hand-drawn or digitally sketched storyboard images would appear here. Each storyboard is accompanied by a caption.

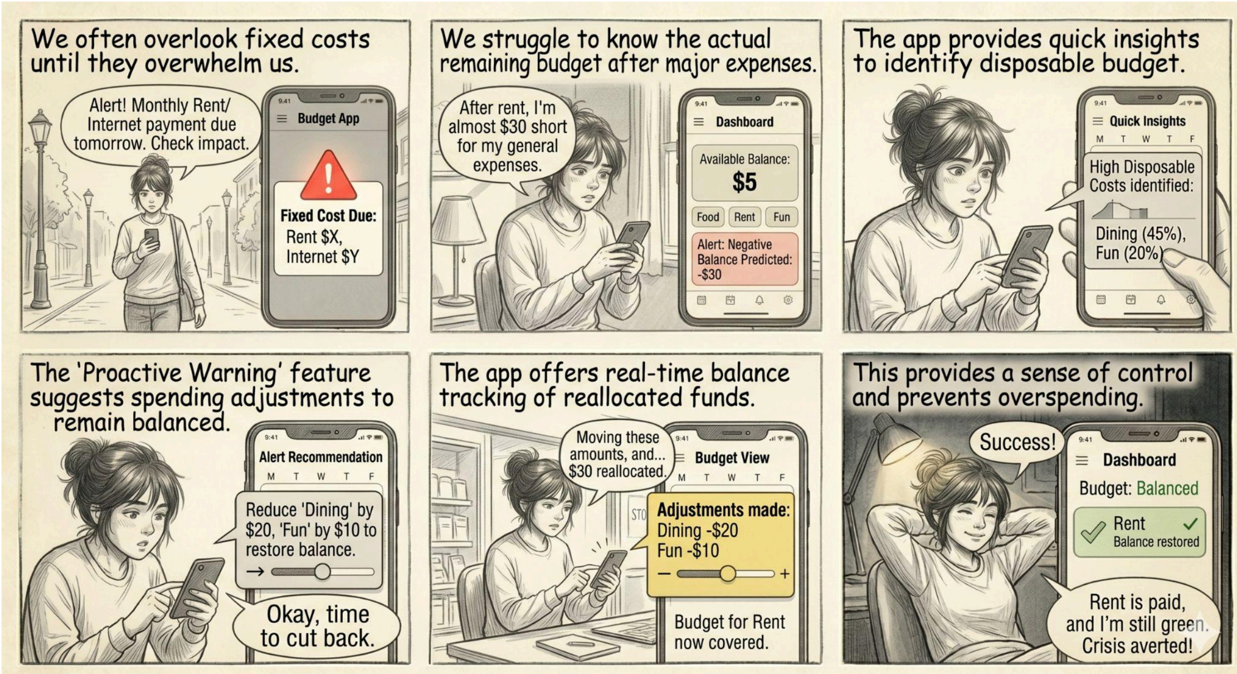
Storyboard 1: Camera Receipt Scan with Verification

Panel	Action	Caption
1	User (P1 archetype) returns from bazaar holding a crumpled handwritten receipt	After shopping, the user has a messy local receipt.
2	Opens DailyKhata and taps the camera icon on the home screen	One tap to start, no login, no menu navigation.

3	App processes the receipt and shows extracted fields (item, amount, date)	AI shows what it understood; uncertain fields are highlighted in yellow.
4	User taps a yellow field and corrects the amount via numpad	Inline edit before saving, user stays in control.
5	User taps “Confirm”	Entry is saved. Home screen shows updated daily total. Expense logged in under 30 seconds.

Storyboard 2: Voice Logging for Informal Cash Expense

Panel	Action	Caption
1	User (P7 archetype) pays rickshaw fare, no receipt given	Informal cash transaction with no paper trail.
2	Opens the app and taps the microphone icon on the home screen	Microphone is one tap away, no navigation needed.
3	Says: “Spent 120 rupees on rickshaw.”	Natural language input; no specific phrasing required.
4	App displays extracted details (Category: Transport, Amount: Rs.120, Note: Rickshaw)	AI parses intent; user sees a summary with a “Confirm/Edit” prompt before saving.
5	User taps “Confirm”	Done. Entire flow takes under 15 seconds with no typing required.



Storyboard 3: Monthly Summary Review

Panel	Action	Caption
1	End of month. User (P6 archetype) wants to understand spending	Monthly review moment, a core use case.
2	Opens DailyKhata and taps “This Month” on home screen	Single tap to monthly view.
3	Shows plain-language summary of spending	“You spent Rs.28,400 this month. Food: 45%, Transport: 20%, Misc: 35%.” Summary in plain text — no chart overload.
4	User taps “Misc” to drill down into entries	Drill-down reveals individual transactions; user identifies untagged items.
5	User re-categorizes two entries	Summary auto-updates. Feedback loop keeps data accurate over time.

Section 3: Personas and Scenarios

Persona 1 – Ammi Jaan (The Homemaker)

Field	Detail
Name	Rukhsana Bibi (Ammi Jaan)
Age	43
Location	Lahore, middle-income residential neighbourhood
Occupation	Full-time homemaker; manages household budget of ~Rs.40,000/month
Digital Literacy	Low, uses WhatsApp for family chat; no app store experience
Devices	Android smartphone (mid-range, 3GB RAM), shared family tablet
Current Practice	Physical Khata notebook; writes daily totals, rarely itemized; receipts kept in a kitchen drawer for reference
Goals	Understand weekly spending; detect irregularities; share simple summaries with husband without complexity
Pain Points	Handwritten receipts fade; forgets small purchases; apps feel "too English" and complicated
Quote	"Main bas ye jaanna chahti hoon ke hafte mein kitna gaya." (I just want to know how

	much was spent in the week.)
Tech Attitude	Distrustful of automation; verifies AI output before accepting; prefers Urdu UI or simple icon-based interaction

Persona 2 - The Hustle Student

Field	Detail
Name	Zaid Malik
Age	22
Location	Lahore, student hostel; monthly stipend + part-time gig income
Occupation	Final-year CS student; occasional freelance graphic design work
Digital Literacy	High, comfortable with apps, spreadsheets, and APIs
Devices	Mid-range Android phone (4GB RAM), laptop for freelance work
Current Practice	Tried Wallet and Cashew but abandoned due to tedious categorization; now uses a Notion table
Goals	Separate personal and freelance expenses for tax purposes; track gig income alongside spending; quick access to financial data
Pain Points	No app handles mixed PKR/foreign-currency freelance income; categorization takes too long; workflow friction
Quote	“I want to spend 5 seconds logging, not 5 minutes.”
Tech Attitude	Trusts AI but wants transparency in reasoning; open to advanced features like Pro View and CSV export

3.3 Scenarios

Scenario 1 (Rukhsana - Camera Flow)

It's Saturday afternoon. Rukhsana has just returned from the weekly bazar with a handwritten receipt from the sabzi wala. She opens DailyKhata, taps the camera icon, and points her phone at the receipt. The app shows her what it read: 'Vegetables - Rs.650.' One number is flagged in yellow because the handwriting was unclear. She taps it, types '850', and confirms. The home screen now shows her weekly total updated. She puts the receipt in the drawer anyway - habit - but feels no anxiety because she knows it's already logged.

Scenario 2 (Zaid - Voice + Pro View Flow)

Zaid is in a rickshaw after a client meeting. He pays Rs.200 cash. While the rickshaw is still moving, he opens DailyKhata and says 'Rickshaw 200 rupees, business.' The app logs it under Transport, tags it as a business expense. Later that evening, he switches to Pro View on his phone, reviews the day's 8 entries, merges two duplicates he spotted, and exports the week's data as a CSV to send to his accountant. Total time: under 2 minutes.

Scenario 3 (Rukhsana - Monthly Summary)

It's the 30th. Rukhsana's husband asks how much was spent on grocery this month. She opens DailyKhata, taps 'This Month', and sees a plain-language line: 'Ration: Rs.18,200.' She shows him the phone. He asks about a Rs.3,000 entry under Misc - she taps it, sees it was the plumber visit, and re-labels it 'Utility'. The summary updates instantly. The entire interaction takes under a minute and requires no explanation of the UI.

Section 4: Competitive Analysis

4.1 Selection Criteria

Competitors were selected if they (a) target personal/household expense tracking, (b) offer mobile apps available in Pakistan, and (c) include at least one of: receipt scanning, voice input, or AI-assisted categorization. We also included common analogue methods as baseline comparators. App Store reviews (Google Play), Reddit discussions, and direct app testing were used to evaluate UX quality.

4.2 Competitive Matrix

Feature / Dimension	Khata Notebook	Bank App	Cashew	Wallet by BudgetBakers	Notion (DIY)	DailyKhata (Proposed)
Receipt Scanning	No	No	Basic OCR	Basic OCR	No	AI (handwritten + mixed

						language)
Voice Input	No	No	No	No	No	Yes (NLP, conversational)
Informal/No-receipt logging	Yes (manual)	No	Manual only	Manual only	Yes (manual)	Yes (voice, one tap)
Urdu/Local receipt support	Yes (native)	No	No	No	N/A	Yes (multilingual OCR)
Verification / error correction	N/A	N/A	Auto-commit	Auto-commit	Manual	Inline edit before save
Local spending categories	User-defined	N/A	Generic (Western)	Generic (Western)	User-defined	Local defaults + custom
Offline functionality	Yes	No	Partial	Partial	Yes	Yes (offline-first)
Summary / analytics	No	Partial	Basic charts	Basic charts	Manual	Plain text + optional charts
Onboarding complexity	None	High	Medium	High	High	Minimal (functional in 60s)
Low-literacy accessibility	Yes	No	No	No	No	Yes (icon-first, Urdu support)
Target User Fit (Pakistan)	High	Low	Low	Low	Medium	High

4.3 Key Findings

Best Practices to Adopt:

- Cashew and Wallet demonstrate that automated categorization is valued, users like not having to think about categories. DailyKhata will adopt this but add local defaults.
- Wallet's 'Smart Receipt' feature (basic OCR) shows willingness to use camera input. DailyKhata extends this to handwritten and mixed-language receipts.
- Notion's flexibility shows power users want raw data access. DailyKhata will offer CSV export and Pro View for this segment.

Opportunities for Differentiation:

- No competitor supports voice logging as a primary input, all treat it as an afterthought or omit it entirely.
- No competitor addresses handwritten or Urdu-English mixed receipts. This is the largest gap for the Pakistani market.
- All digital competitors have high onboarding friction and abandon low-literacy users immediately. A minimal, icon-forward onboarding is an open opportunity.
- Verification after AI parsing is absent in all competitors, DailyKhata's inline edit step directly addresses the trust deficit identified in interviews.
- No competitor is designed offline-first. In areas with spotty connectivity, this is a significant practical differentiator.

Appendix: Needs Finding Study Plan (from Milestone 1)

The following is the Needs Finding Study Plan submitted in Milestone 1, reproduced here for reference. No changes were made to the study plan from Milestone 1; the protocol was followed as designed.

High-Level Study Goals

The primary goal is to understand how Pakistani homemakers and students currently track daily expenses across informal and formal retail channels. We aim to identify the specific friction points that cause them to abandon financial tracking, and observe their mental models regarding how they categorize, verify, and store physical receipts.

Recruiting Criteria and Strategy

We recruited using convenience and snowball sampling. Criteria: Group A (homemakers/local shop frequenters, primarily analogue methods). Group B (university students/freelancers who have attempted digital finance apps). All participants must actively manage a monthly budget and regularly receive physical receipts. Interviews lasted 30–45 minutes and included workflow observations.

Interview Protocol

Overarching Question: How do users across different digital literacy levels currently track, manage, and abandon their daily expense logs, particularly when dealing with physical or informal receipts?

Introduction

Hi, we're Saad and Khadijah, here to understand how you manage daily expenses. We're designing an app to make financial tracking frictionless. The interview will take about 10–20 minutes and is completely voluntary. Comments will be anonymized. May we record the audio?

Warm-up:

- Tell us about your typical week, who handles shopping and expense tracking in your household?
- Follow-up: What tools do you currently use to keep track of money?
- Follow-up: Are you satisfied with your current approach?
- Follow-up: What's the most frustrating part of managing money?

Recent Concrete Instance, The Shopping Trip:

- Think back to your last grocery run. What happened after you paid?
- Follow-up: Did you get a receipt? Was it printed or handwritten?
- Follow-up: Where did you put that receipt when you got home?
- Follow-up: Did you write the expense down? When and how?

Workflow Observation:

- Can you show us the notebook, app, or system you use and walk us through your process?
- Observation: If you had a receipt for 'milk and bread' from a local dairy right now, show us how you would record it.
- Follow-up: How do you categorize things?
- Follow-up: How much time per week does this take?

Friction and Abandonment:

- Think of a time you wanted to record an expense but didn't. What stopped you?
- Follow-up: Was it a lost receipt or too much effort?
- Follow-up: How do you handle no-receipt cash expenses (e.g., rickshaw)?
- Follow-up: Have you tried an expense-tracking app? If you quit, why?

Concept Introduction:

- If you could take a photo of a receipt, or say 'I spent 500 on vegetables,' how would that changebyour routine?
- Follow-up: Would you trust an AI to read a handwritten receipt correctly?
- Follow-up: How should the app behave if it's unsure about a number?
- Follow-up: Would you prefer typing, speaking, or snapping photos?

Conclusion

Thank you for your time. Your insights are invaluable. Is there anything else about managing expenses we haven't covered? Feel free to reach out if anything comes to mind later.